

Pr. 3 Spread Locator

1) What is Statistical distributions?

ans. Statistical distribution shows how data values are spread or occur in a dataset.

data = { 50, 60, 60, 70, 70, 70, 80 }.

70 occurs most often, so the distribution tells us how frequently different marks appear.

2 What is Q-Q Plot and why is it used?

ans. Quantile - Quantile Plot is graph used to check whether data follows a particular distribution, especially a normal distribution.

Example: If the points in a Q-Q Plot are close to a straight diagonal line, the data is approximately normal distributed.

3 Difference between Discrete and continuous Distribution?

ans Discrete countable values

Example: 2, 3, 4, 5, 6

Continuous Kisi range me koi bhi Value.

Example: Transaction amount = 325475

4) What is Bernoulli distribution?

ans Bernoulli means possible outcome only 2 Success / Failure.

5) What is Binomial distribution?

ans Binomial distribution multiple Bernoulli trials me Successes ki number ko represent karta ha.

Example: 10 Transactions me kitni Successful hua.

6 Explain Log - Normal distribution?
Ans Jab kisi variable ka log approximately normally distributed ho, usse Log-normal distribution kehte hain.

7 Explain Power Law distribution?
Ans Power Law me few values bahut large hoti hain aur majority values small hoti hain.

Example : Few customers ke transaction amounts bahut high honge.

8 What is Box - Cox Transform?
Ans Box - Cox, a mathematical transformation hai jo skewness reduce or data ko normal banane me help karte ha.

9 Explain Poisson distribution with an example?

ans Poisson distribution Fixed time / Space me events ki Count ko model karta hai.

Example :- If a Shop receives an average of 7 Customer transactions per hour, Poisson distribution can be used to find the probability of receiving 10 transactions in the next hour.

Poisson :- Fixed time / Space mein event kitni baar hai?

10 What is Z-Score Probability?

ans Z-Score batata hai ki koi value mein se kitni Standard deviations door hai.

Example :- $Z=2$ means value mean se 2 Standard deviation above hai.

11 Difference between PDF and CDF ?
ans PDF Probability density Show Karta
hai

Example: Kisi value / range ke around
density

CDF
Cumulative Probability Show Karta hai.

Example: Kisi value tak total
Probability.

PDF : Probability ki density
CDF : Probability ka total / cumulative
amount.

The End.